

BIG TROUT LAKE, ON, Canada

WMO: 716750

Lat: 53.820N

Lon: 89.900W

Elev: 223

StdP: 98.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.6	-34.2	-40.9	0.1	-36.2	-38.5	0.1	-34.0	8.9	-21.3	8.1	-19.7	2.2	290	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	26.9	18.5	25.2	17.6	23.5	16.8	20.1	24.6	18.9	23.2	17.9	21.9	3.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.8	22.1	17.3	12.7	20.8	16.2	11.8	20.0	59.1	24.9	54.9	23.2	51.4	21.8	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.7	(b) 7.7	(c) 6.9	(e) -40.5	(f) 30.7	(g) 3.0	(h) 1.9	(i) -42.6	(j) 32.1	(k) -44.4	(l) 33.2	(m) -46.1	(n) 34.2	(o) -48.2	(p) 35.6		
(a) 8.7	(b) 7.7	(c) 6.9	(e) -40.3	(f) 22.6	(g) 2.8	(h) 1.7	(i) -42.3	(j) 23.8	(k) -44.0	(l) 24.7	(m) -45.6	(n) 25.7	(o) -47.7	(p) 26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.5	-21.7	-19.8	-12.0	-2.8	5.6	13.0	16.4	15.0	9.8	2.3	-7.8	-17.6
	DBStd	14.75	7.56	7.37	8.16	6.77	5.32	4.80	3.69	4.00	4.67	4.60	6.87	8.13
	HDD10.0	4764	984	836	683	384	156	22	2	4	61	243	534	856
	HDD18.3	7303	1242	1069	941	633	393	169	83	118	258	498	784	1114
	CDD10.0	553	0	0	0	1	21	113	199	160	54	4	0	0
	CDD18.3	51	0	0	0	0	0	11	23	16	2	0	0	0
	CDH23.3	398	0	0	0	0	11	105	177	96	10	0	0	0
	CDH26.7	61	0	0	0	0	1	17	28	13	1	0	0	0
Wind	WSAvg	3.4	3.2	3.2	3.2	3.1	3.1	3.1	3.4	3.7	4.0	3.8	3.3	
Precipitation	PrecAvg	570	21	16	26	28	49	62	80	85	73	48	40	27
	PrecMax	705	42	32	50	65	130	135	149	155	168	99	61	66
	PrecMin	304	9	7	4	1	2	16	11	24	13	9	19	6
	PrecStd	98	8	7	11	18	33	28	43	38	46	21	13	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.5	1.6	9.5	17.6	24.6	28.6	29.6	28.4	24.8	17.0	8.5	1.1
		MCWB	-1.1	0.6	5.2	9.6	14.4	18.6	20.8	20.1	17.6	13.6	6.5	0.0
	2%	DB	-3.3	-1.0	5.6	13.2	21.1	26.1	27.0	25.9	21.7	13.9	4.9	-1.6
		MCWB	-3.8	-2.0	2.1	6.8	12.6	16.9	18.7	19.1	16.0	10.7	3.3	-2.1
	5%	DB	-6.1	-3.8	3.0	10.0	18.1	24.1	25.3	23.9	19.3	11.5	2.7	-3.6
		MCWB	-6.5	-4.7	0.7	4.5	11.3	16.1	17.9	17.8	15.0	9.2	1.6	-4.1
	10%	DB	-9.7	-7.6	0.5	7.1	15.3	21.8	23.5	21.9	17.2	9.1	0.9	-5.7
		MCWB	-10.1	-8.5	-1.1	2.9	9.6	15.1	17.1	16.9	14.0	7.3	0.2	-6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	0.8	5.7	10.3	16.1	20.8	22.6	21.5	19.0	14.5	6.6	0.1
		MCDB	-0.5	1.7	9.2	16.2	21.6	24.8	27.1	26.0	22.4	17.0	8.2	0.7
	2%	WB	-3.8	-1.9	2.6	7.1	13.7	18.6	20.4	19.9	17.2	11.3	3.6	-2.1
		MCDB	-3.3	-0.9	5.0	12.6	19.6	23.5	24.6	24.4	20.2	13.2	4.5	-1.7
	5%	WB	-6.5	-4.8	0.8	5.0	11.9	17.1	19.0	18.7	15.6	9.4	1.6	-4.1
		MCDB	-6.1	-3.8	2.9	9.4	16.9	21.9	23.4	22.7	18.6	11.4	2.5	-3.7
	10%	WB	-10.1	-8.5	-1.0	3.2	10.0	15.7	18.0	17.6	14.1	7.3	0.2	-6.1
		MCDB	-9.7	-7.6	0.8	6.9	14.6	20.3	22.1	21.0	16.8	9.2	0.9	-5.7
Mean Daily Temperature Range	5% DB	MDBR	9.7	12.1	13.4	12.8	12.0	11.9	11.2	10.2	8.7	6.0	6.4	8.1
		MCDBR	10.6	12.1	13.1	15.8	16.9	14.2	13.0	11.9	11.7	9.8	6.6	7.9
		MCWBR	10.5	11.1	10.4	10.3	9.8	7.0	6.6	6.5	7.1	7.0	5.5	7.6
	5% WB	MCDBR	10.6	11.8	12.1	14.5	15.3	12.3	11.3	10.9	9.9	9.2	6.2	8.1
		MCWBR	10.5	11.0	10.0	9.9	9.4	7.2	6.5	6.5	6.7	7.0	5.4	7.9
Clear-Sky Solar Irradiance	taub	0.240	0.252	0.269	0.336	0.355	0.355	0.388	0.361	0.331	0.291	0.263	0.225	
	taud	2.236	2.278	2.307	2.304	2.402	2.442	2.382	2.465	2.522	2.576	2.376	2.258	
	Ebn at Noon	743	863	924	893	892	894	854	861	849	820	716	695	
	Edh at Noon	59	80	97	114	109	106	111	97	80	61	52	47	
All-Sky Solar Radiation	RadAvg	1.03	2.23	3.76	5.27	5.35	5.73	5.38	4.35	2.89	1.60	0.79	0.66	
	RadStd	0.10	0.14	0.28	0.37	0.47	0.60	0.52	0.38	0.36	0.16	0.10	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	-0.52	-0.74	-0.60	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	-0.52	-0.74	-0.60	N/A	N/A	N/A	N/A

Nomenclature: See separate page